

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional requirement(Epic)	User Story Number	User Story/Task	Story Points	Priority	Team Member	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Preparation	USN-1	As a developer, I want to collect and preprocess the agricultural dataset for model training.	3	High	Dharshan Sai Kalavakunta	26 June 2026	30 June 2026
Sprint-1	Feature Engineering	USN-2	As a developer, I want to prepare input features (N, P, K, Temperature, Humidity, pH, Rainfall).	2	High	Dharshan Sai Kalavakunta	26 June 2026	30 June 2026
Sprint-2	Machine Learning	USN-3	As a developer, I want to train and compare multiple machine learning models.	5	High	Dharshan Sai Kalavakunta	26 June 2026	01 July 2026
Sprint-2	Model Deployment	USN-4	As a developer, I want to save the best-performing model as a Pickle (.pkl) file.	2	High	Dharshan Sai Kalavakunta	27 June 2026	01 July 2026

Sprint-3	Backend Development	USN-5	As a user, I want to submit agricultural parameters through the Flask application to receive crop recommendations.	5	High	Dharshan Sai Kalavakunta	27 June 2026	02 July 2026
Sprint-3	Frontend Development	USN-6	As a user, I want a responsive interface to enter parameters and view prediction results.	3	High	Dharshan Sai Kalavakunta	27 June 2026	02 July 2026
Sprint-4	Testing	USN-7	As a developer, I want to test and validate prediction accuracy and application functionality.	3	High	Dharshan Sai Kalavakunta	29 June 2026	03 July 2026
Sprint-4	Documentation	USN-8	As a developer, I want to prepare documentation and finalize the project for submission.	2	High	Dharshan Sai Kalavakunta	29 June 2026	03 July 2026

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start date	Sprint End Date(Planned)	Story Points Completed	Sprint Release Date(actual)
Sprint-1	5	5 Days	26 June 2026	30 June 2026	5	30 June 2026
Sprint-2	7	6 Days	26 June 2025	01 July 2026	7	01 July 2026
Sprint-3	8	6 Days	27 June 2026	02 July 2026	8	02 July 2026
Sprint-4	5	5 Days	29 June 2026	03 July 2026	5	03 July 2026

Velocity: Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \text{sprint duration} / \text{velocity} = 25/4 = 6.25$$

6 Story Points per Sprint